

UPDATED ARCHITECTURE

GuardianAI

Architecture Evolution: Checkpoint-1 → Checkpoint-2

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1. Purpose of This Document

At Checkpoint-1, the team submitted an initial architecture sketch as part of the SRS (a single Mermaid diagram in Section 2.1). This document records how that architecture has evolved during the Checkpoint-2 build, **why** each change was made, and what the resulting system looks like now that it is running. It is a companion to, not a replacement for, the full Software Design Document (SDD v1.0) — this document focuses on the *delta* and its rationale; the SDD contains the complete, exhaustive design (component design, class model, data design, algorithms) for the as-built system.

2. Checkpoint-1 Architecture (Initial Sketch)

Figure 1 reproduces the architecture proposed at Checkpoint-1, redrawn from the Mermaid source embedded in SRS v2. It was a design intent, written before the corresponding code existed — several labels (YOLOv8-Pose, "Hybrid Decision Engine") named the intended technology, but at that stage the running prototype still used MediaPipe and rule-based detection only, with no dashboard.

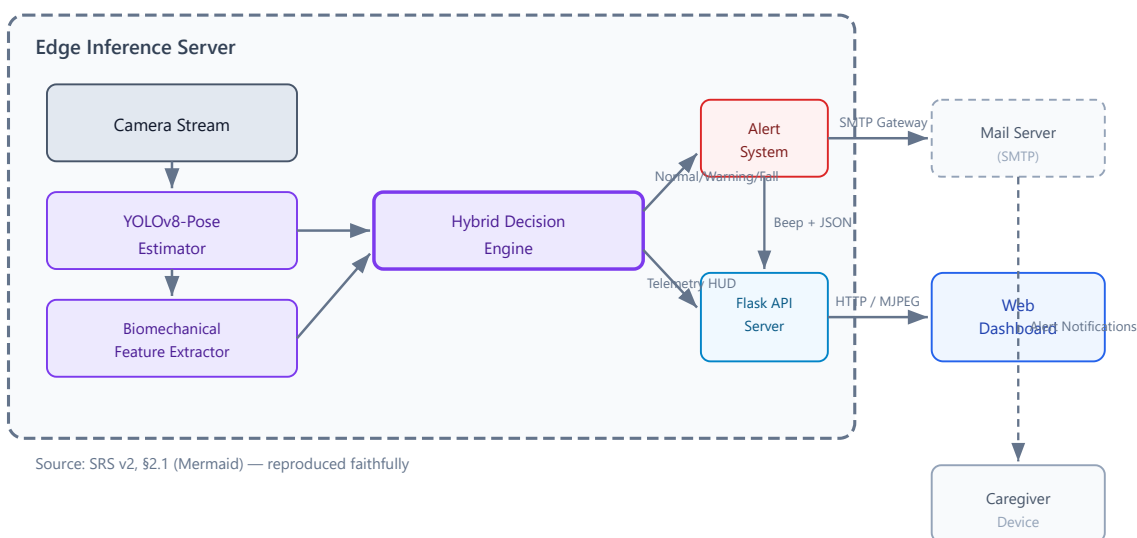


Figure 1 — Checkpoint-1 architecture sketch (design intent, pre-implementation).

2.1 What the Checkpoint-1 sketch did not specify

Being a proposal-stage diagram, it named components without defining their behaviour. Several things a working system needs were entirely absent:

- No fallback behaviour if the neural model or camera failed.
- No operator-facing controls of any kind (pause, mute, reset).
- Only two notification channels shown (beep, SMTP) — no on-screen popup, no native OS notification.
- No concurrency model — nothing said whether video streaming and the REST API could run simultaneously.
- No indication of which pose technology would actually ship — the codebase at this stage still used MediaPipe.

3. Checkpoint-2 Architecture (As Built)

Figure 2 shows the current, verified architecture. Every element shown has been exercised end-to-end against a live webcam feed. Full component-level detail (class responsibilities, data schemas, algorithms) is documented in SDD v1.0, Sections 3–9; this diagram summarises the same system at the level needed to see what changed.

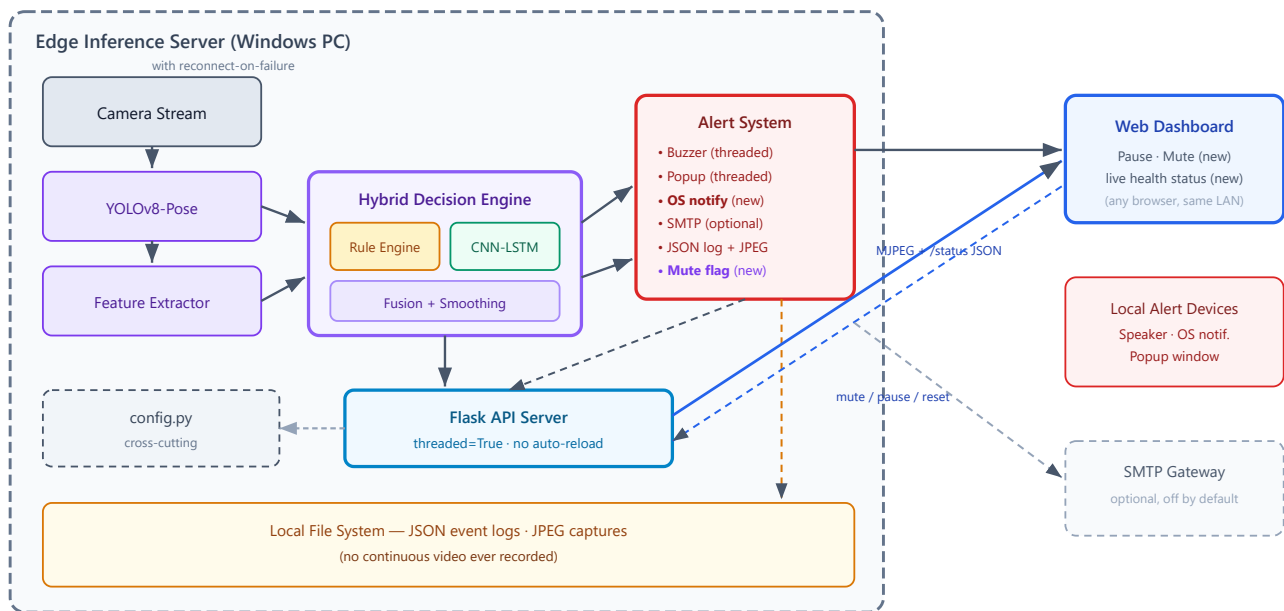


Figure 2 — Checkpoint-2 architecture (as built and verified). Bold-labelled items inside the Alert System box are additions not present in the Checkpoint-1 sketch.

4. What Changed, and Why

4.1 Pose estimation technology CHANGED

	Checkpoint-1	Checkpoint-2 (current)
Technology	MediaPipe (running code) / YOLOv8-Pose (named in the sketch only)	YOLOv8-Pose (nano) — actually implemented and verified
Rationale	MediaPipe is fundamentally single-person oriented and the target deployment (a ward or shared living space) routinely contains several people. YOLOv8-Pose detects persons and regresses keypoints in one pass, is more robust to partial occlusion (furniture, other people), and gives a credible path to the multi-person tracking already scoped in FR-1.6.	

4.2 Hybrid decision engine REALISED

The Checkpoint-1 sketch *named* a "Hybrid Decision Engine" as a single box. It is now a concrete, working sub-architecture: an explainable rule-based state machine and a CNN-LSTM temporal classifier run independently every frame, and their verdicts are fused with a severity-priority override (rules win when they

detect greater danger with high confidence). This satisfies the explainability constraint identified in the SRS (NFR-S.2) that a purely black-box verdict would not.

4.3 Alert subsystem EXPANDED

Channel	Checkpoint-1	Checkpoint-2	Why added
Console output	—	Added	Baseline auditability during development.
Audible buzzer	Named ("WinBeep")	Implemented, threaded	Original design would have blocked the frame loop for ~600 ms per alert; moved to a daemon thread.
On-screen popup	—	Added, threaded, duration configurable	A single audible beep is easy to miss; a visible high-priority window reinforces it.
Native OS notification	—	Added	None of the Checkpoint-1 channels were visible if the caregiver had the dashboard closed or was working in another application. This closes that gap directly.
SMTP e-mail	Named	Implemented, off by default	Retained as designed; disabled until credentials are supplied so it cannot fail silently in the background.
JSON logging + JPEG capture	Named	Implemented, with cooldown and log continuity across restarts	Matches original intent; hardened so a restart mid-day does not truncate the log.

4.4 Operator control NEW

The Checkpoint-1 architecture had no notion of an operator interacting with a running system — it was alert-out only. Three controls were added directly in response to running the prototype live:

- **Pause / Resume feed** — stops frame processing entirely (and therefore all alerts), for demonstrations where the audible buzzer becomes disruptive.
- **Mute / Unmute alerts** — a finer-grained control than Pause: suppresses buzzer, popup and OS notification while video, telemetry and event logging continue uninterrupted, so the evidentiary record is never affected by muting.
- **Reset tracking** — clears the feature buffer and smoothing history without restarting the process.

4.5 Reliability and concurrency NEW

The Checkpoint-1 sketch made no statement about failure handling or concurrency. Two defects surfaced only once the system was actually run continuously, and both are now part of the architecture:

- **Camera reconnection.** The initial video route ended the stream permanently on the first failed frame read. It now retries briefly, then performs a full reopen with backoff, and always releases the device handle on client disconnect.

- **Concurrent request handling.** The Flask development server defaults to one request at a time, which meant the long-lived video stream could starve the telemetry API. The server now runs with `threaded=True`, and the auto-reloader — which was found to restart the whole process spuriously and tear down the camera mid-stream — is disabled.

4.6 Live system-health reporting CHANGED

Not part of the original sketch at all, but worth recording as an architectural correction: the dashboard's "System" and "Detection" indicators were initially static markup that read "Online" / "Active" unconditionally — including with the camera disconnected. The `/status` endpoint now reports real `camera_connected` and `alerts_muted` state, and the dashboard renders from it.

5. Summary of Architectural Deltas

Aspect	Checkpoint-1	Checkpoint-2	Type
Pose estimation	MediaPipe (actual) / YOLOv8 (planned)	YOLOv8-Pose (actual)	Changed
Decision logic	Named concept	Rules + CNN-LSTM, fused	Realised
Alert channels	2 (beep, SMTP)	5 (+ console, popup, OS notify)	Expanded
Operator controls	None	Pause, Mute, Reset	New
Camera fault handling	Unspecified	Retry + reopen + clean release	New
Concurrency model	Unspecified	Threaded server, no auto-reload	New
Health reporting	Unspecified	Live, state-derived indicators	Changed
Web dashboard	Named box	Full implementation, verified	Realised

6. Forward Reference

This document intentionally stops at the level of components and their interactions. For the complete design — class responsibilities, data schemas, sequence diagrams, algorithm pseudocode, non-functional design and the requirements traceability matrix — see **Software Design Document v1.0**

(docs/SDS_GuardianAI_v1.0.pdf), Sections 3 through 12. Known limitations in the current build, including an open defect in the fall-confirmation rule, are recorded honestly in SDD §12.2 and carried into SRS v3 §9.4 rather than papered over in either document.

End of Updated Architecture Document

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